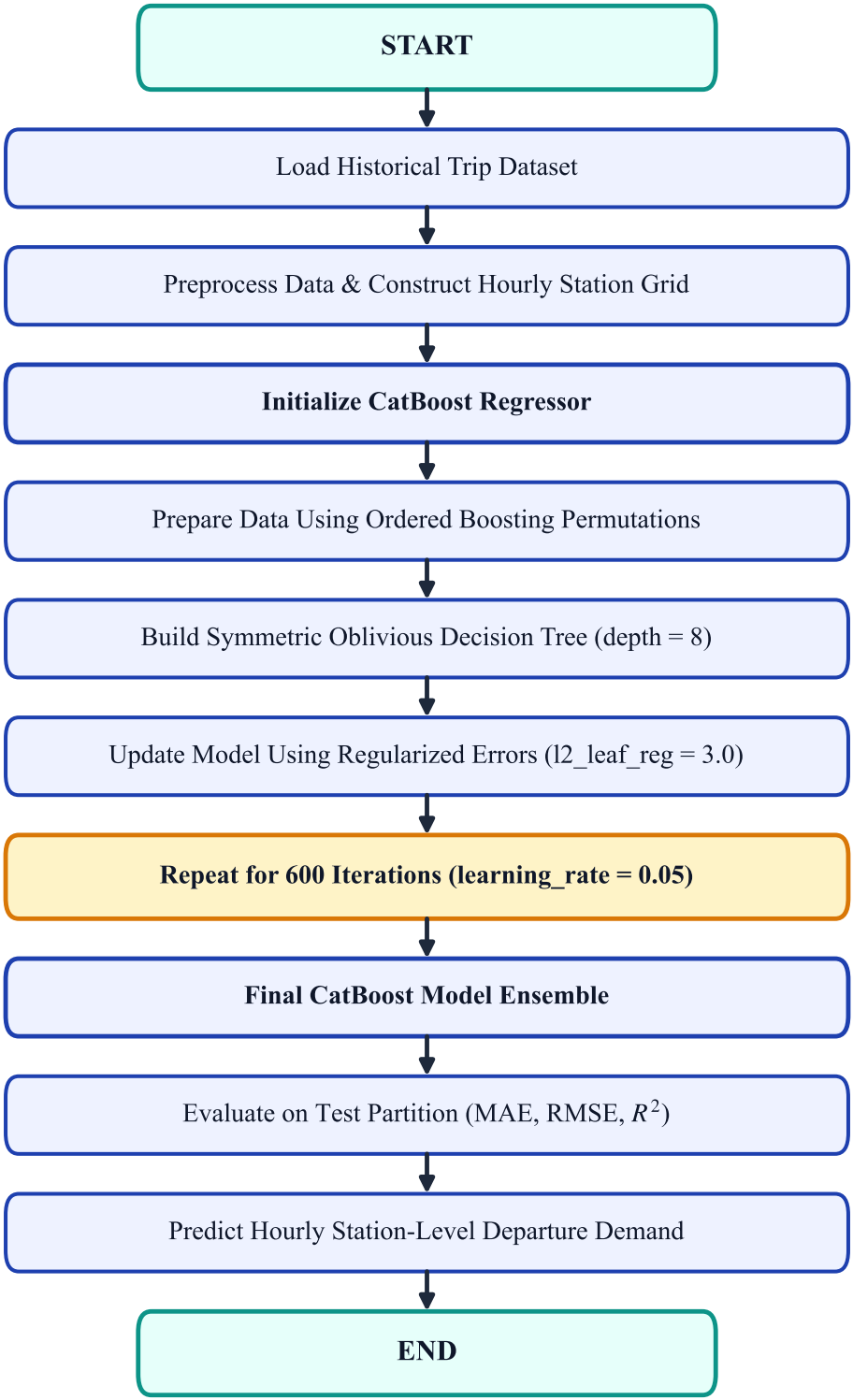
START
Load Historical Trip Dataset
Preprocess Data & Construct Hourly Station Grid
Initialize CatBoost Regressor
Prepare Data Using Ordered Boosting Permutations
Build Symmetric Oblivious Decision Tree (depth = 8)
Update Model Using Regularized Errors (l2_leaf_reg = 3.0)
Repeat for 600 Iterations (learning_rate = 0.05)
Final CatBoost Model Ensemble
Evaluate on Test Partition (MAE, RMSE, R^2)
Predict Hourly Station-Level Departure Demand
END